FORM TP 2022016

TEST CODE **01229020**

JANUARY 2022

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
5. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

1. (a) Define EACH of the following terms as it relates to data:

(i) Verification

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(2 marks)

(ii) Validation

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(2 marks)

(b) Outline the function of EACH of the following validation methods. Give ONE example of EACH method.

(i) Range check

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Example .....

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(2 marks)

(ii) Inconsistency check

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Example .....

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(2 marks)

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(c) The spreadsheet below shows the runs scored, balls faced and the strike rate of each player who contributed to the total score of 199 runs for the Warriors Cricket Team in a T20 game.

The strike rate of each player is calculated as runs scored divided by balls faced multiplied by 100.

The percentage of total runs scored for each player is calculated as the runs scored by that player divided by the total runs scored.

	A	B	C	D	E
1	Warriors Scorecard				
2	Player Name	Runs Scored	Balls Faced	Strike Rate	% Runs Scored
3	Alphonso	18	28	64.29	9.05
4	Callender	58	26	223.08	29.15
5	Forde	7	9	77.78	3.52
6	Alleyne	43	22	195.45	21.61
7	St Louis	44	25	176.00	22.11
8	Kellar	21	10	210.00	10.55
9	Extras	8		#DIV/0!	4.02
10	Total	199	120	165.83	100.00

A formula was inserted in Cell D3 to calculate the strike rate of the player, Alphonso, who scored 18 runs and faced 28 balls. This formula was also copied to the range of Cells D4 to D10.

(i) Write the formula which was inserted in Cell D3.

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(2 marks)

(ii) Outline what accounted for the result displayed in Cell D9, after copying the formula from Cell D3 to the range of Cells D4 to D10.

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(2 marks)

GO ON TO THE NEXT PAGE

(d) State how EACH of the following types of data is displayed within a cell, in an electronic spreadsheet:

(i) Numeric data

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(1 mark)

(ii) Alphanumeric or text data

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(1 mark)

(e) Write the formula inserted in Cell E3 before it was copied to Cells E4 to E10.

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(2 marks)

(f) State the type of cell referencing used in the formula in EACH of the following cells:

(i) Cell D3 .....

(ii) Cell E3 .....

(2 marks)

(iii) Explain why the type of cell referencing stated in (f) (ii) is used in Cell E3.

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(2 marks)

Total 20 marks

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2. (a) State TWO methods that can be used to prevent members of the public from having physical access to stored information.

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(2 marks)

(b) Suggest TWO software methods that can be used to prevent unauthorized staff from accessing sensitive data on a network.

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(2 marks)

(c) State TWO ways in which the integrity of data can be destroyed.

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(2 marks)

(d) Cyndra has created a web-enabled database application to allow her friends from the university to enter their own data from a webpage. The database consists of the following two tables, CONTACTS and DEGREES. Data entered in the Gender field must be "M" for male and "F" for female and all dates of birth entered must be prior to 30 June 2000.

TABLE 1: CONTACTS DATABASE

Contacts							
ID	**FirstName**	**LastName**	**BirthDate**	**Gender**	**Married**	**Phone**	**Degree**
1001	Roy	CLARKE	12/01/1997	M	Yes	688-4444	010
1002	James	JOHNSON	22/03/1982	F	No	720-9123	100
1003	Tyson	LENNY	23/03/1996	M	Yes	630-1234	030
1004	Dennis	DRAKES	17/04/1978	M	Yes	456-7890	050
1005	Michael	BAPTISE	25/12/1999	M	No	565-6676	030

TABLE 2: DEGREES DATABASE

Degrees	
Degree Code	**Degree Programme**
010	BSc Economics
030	BSc Computer Science
050	BA History
100	BSc Mathematics

(i) Identify the MOST appropriate field types for EACH of the following fields.

Field Name	Data Type
BirthDate	
Married	
Degree	
Gender	

(4 marks)

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(ii) State what is meant by the term 'primary key' in a database table.

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(2 marks)

(iii) Identify TWO primary keys from the database tables **on page 9**.

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(2 marks)

(iv) Outline the role of a 'foreign key' in a database table.

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(2 marks)

(v) Identify the field(s) from the database tables mentioned **on page 9** that can be used to establish EACH of the following foreign keys.

Table Name	**Field(s)**
CONTACTS	
DEGREES	

(2 marks)

(vi) Describe the relationship that is established between Table 1 and Table 2 **on page 9**.

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(2 marks)

Total 20 marks

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3. (a) Roger, a new employee in the information technology department of a supermarket, notices that discounts are being calculated manually. Roger asks for a pseudocode to be generated that would handle 25% discounts on merchandise totals.

Write the pseudocode for the algorithm that will

- initialize the variables
- prompt the user to input the price and quantity of the items
- calculate the discount amount at 25% of the total if the total is greater than $1800
- output the discount amount.

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(10 marks)

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(b) Roger has been tasked with reconfiguring the network system for the supermarket, which is expanding.

(i) List THREE types of networks.

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(3 marks)

(ii) Describe ONE type of network that would be suitable for effectively managing the functions of all departments of the supermarket from a central location.

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(2 marks)

(c) Roger is considering segregating the users of the network. Differentiate between an 'intranet' and an 'extranet'.

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(2 marks)

(d) Roger must produce a table showing the cost of the equipment he will need to purchase in order to facilitate the expansion of the network. The table below illustrates a part of the spreadsheet that was produced.

	A	B	C	D
1	New Network			
2	Item	Number	Cost	Total
3	Router	2	400	800
4	System unit	5	200	
5	Switch	10	250	
6	Cabling	8	80	
7	Storage	5	50	
8	Firewall	5	65	
9	Dock	6	235	
10				
11	Total cost			
12	Average cost			

(i) Write the formula to be inserted in Cell D5 to calculate the total cost of the switches.

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(1 mark)

(ii) Write the function that would be placed in Cell D12 to calculate the average cost.

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(2 marks)

(e) Roger's employer wants to have the spreadsheet **on page 14** sorted in alphabetical order. List, in the correct order, the steps that would be required to do this.

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(3 marks)

(f) During a staff meeting, management decided that docks were an unnecessary cost. State, in the correct order, the steps required to delete Row 9 from the spreadsheet **on page 14**.

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(2 marks)

Total 25 marks

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4. (a) Keva has just graduated from college with a degree in computer science. She decides to set up a home business offering programming services.

(i) List FOUR **main** computer hardware devices that Keva will need to facilitate her business.

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(4 marks)

(ii) Keva has to ensure that her devices are equipped with sufficient storage. Describe the function of any TWO features of a hard drive/disk.

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(4 marks)

(b) Keva is aware of some of the threats to a computer system which may impact the operations of the business.

(i) State ONE reason for using passwords.

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(ii) Give ONE example of an effective password.

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(2 marks)

(c) Keva writes the pseudocode below for her first client. Study the pseudocode carefully and answer the questions that follow.

```
Start
    Line 1.   Set MAX ← 0
    Line 2.   Write "Enter Integer"
    Line 3.   Read NUM
    Line 4.       WHILE NUM <> 0 DO
    Line 5.           IF NUM > MAX THEN
    Line 6.               Set MAX ← NUM
    Line 7.           ENDIF
    Line 8.               Write "enter integer or 0 to stop"
    Line 9.               Read NUM
    Line 10.      ENDWHILE
    Line 11.      Write "max" , MAX
Stop
```

(i) State the purpose of the pseudocode.

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(1 mark)

(ii) Name the constructs that are in the pseudocode.

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(3 marks)

(iii) State the line which represents initialization.

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(1 mark)

(iv) State the purpose of the second output statement.

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(1 mark)

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(v) State the names of the variables that are used in the pseudocode **on page 17**.

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(2 marks)

(vi) Identify the TWO lines in the pseudocode **on page 17** that demonstrate relational operators.

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(2 marks)

(d) Keva, having written the pseudocode in (b) **on page 17**, wants to use flowcharting techniques to show the structure of the program.

Draw a flowchart depicting the WHILE loop.

(5 marks)

Total 25 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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